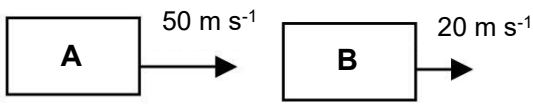
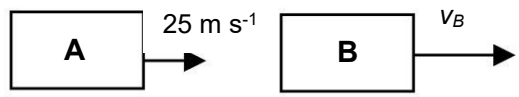
**3.** A 900 kg car A travelling at 50 m $s^{-1}$ collides inelastically with a 1500 kg car B travelling at 20 m $s^{-1}$ in the same direction as shown in Fig. 3.1. Immediately after the collision, the velocity of car A is 25 m $s^{-1}$ and the velocity of car B is $v_B$ as shown in Fig. 3.2.

The collision begins at time *t*. During the collision, both cars are in contact for 120 ms.




**Before**

**Fig. 3.1**




**After**

**Fig. 3.2**

**(a)** **(i)** Determine the magnitude of the velocity $v_B$.

magnitude of velocity $v_B$ = ..... m $s^{-1}$ [1]

**(ii)** On Fig. 3.3, sketch and label the momentum-time graphs for car A and car B, before, during and after the collision. Include numerical values.

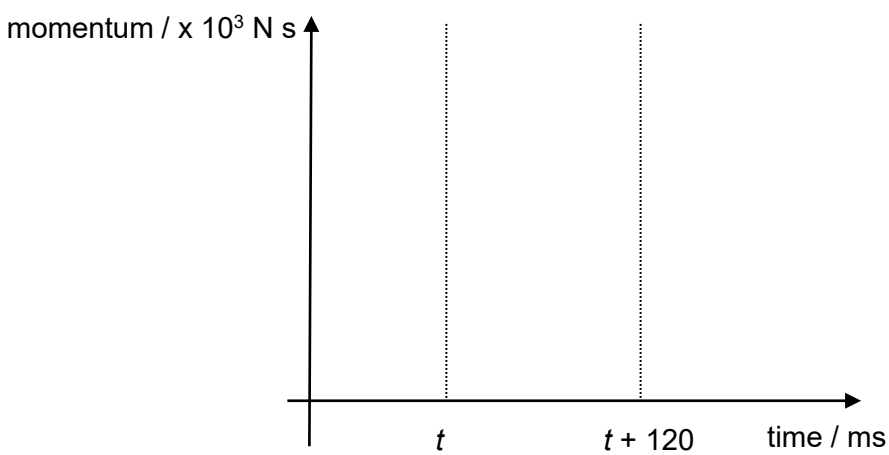



**Fig. 3.3**

[4]

**(b)** Determine the average force that car B exerts on car A during the collision.

average force = ..... N [1]

**(c)** State and explain one design or feature of the car that reduces the force experienced by the driver.

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.....

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..... [2]